

Day 2: Conditional Statements

Topics: if, if-else, nested if, switch

1. To Check Given Number is Even or Odd

Uses the modulo operator (%) to check if a number is divisible by 2.

```
#include <stdio.h>

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n % 2 == 0)
        printf("%d is Even\n", n);
    else
        printf("%d is Odd\n", n);
    return 0;
}
```

2. To Print Largest from Given Three Numbers

Compares three numbers using nested if-else to find the largest.

```
#include <stdio.h>

int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a >= b && a >= c)
        printf("Largest = %d\n", a);
    else if (b >= a && b >= c)
        printf("Largest = %d\n", b);
    else
        printf("Largest = %d\n", c);
    return 0;
}
```

3. To Check Given Year is Leap Year or Not

A year is a leap year if divisible by 4, but century years must be divisible by 400.

```
#include <stdio.h>

int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
        printf("%d is a Leap Year\n", year);
    else
        printf("%d is NOT a Leap Year\n", year);
    return 0;
}
```

4. To Design Student's Grade Calculator

Takes marks as input and assigns a grade based on score ranges using if-else ladder.

```
#include <stdio.h>

int main() {
    int marks;
    printf("Enter marks (0-100): ");
    scanf("%d", &marks);
    if (marks >= 90)
        printf("Grade: A (Excellent)\n");
    else if (marks >= 75)
        printf("Grade: B (Good)\n");
    else if (marks >= 60)
        printf("Grade: C (Average)\n");
    else if (marks >= 40)
        printf("Grade: D (Pass)\n");
    else
        printf("Grade: F (Fail)\n");
    return 0;
}
```

5. To Check Given Number is Positive / Negative / Zero

Uses if-else if-else to classify a number as positive, negative, or zero.

```
#include <stdio.h>

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n > 0)
        printf("%d is Positive\n", n);
    else if (n < 0)
        printf("%d is Negative\n", n);
    else
        printf("The number is Zero\n");
    return 0;
}
```

6. To Design Simple Calculator using Switch Statement

Performs basic arithmetic (+, -, *, /) based on the operator entered by user.

```
#include <stdio.h>

int main() {
    float a, b, result;
    char op;
    printf("Enter first number: ");
    scanf("%f", &a);
    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &op);
    printf("Enter second number: ");
    scanf("%f", &b);
```

```
switch (op) {
    case '+':
        result = a + b;
        printf("Result = %.2f\n", result);
        break;
    case '-':
        result = a - b;
        printf("Result = %.2f\n", result);
        break;
    case '*':
        result = a * b;
        printf("Result = %.2f\n", result);
        break;
    case '/':
        if (b != 0)
            printf("Result = %.2f\n", a / b);
        else
            printf("Error: Division by zero!\n");
        break;
    default:
        printf("Invalid Operator!\n");
}
return 0;
}
```

— End of Day 2 Assignment —